Set-Theoretic Structure and Combinatorial Properties of the 6-Dimensional Hypercube

—— Spin Classification, Set Enumeration, and Sign Product ——

Author: Noriaki Kihara (WF System Co., Ltd.) **Date**: April 2026 **DOI**: 10.5281/zenodo.19646652

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0. Purpose of This Paper

This paper defines a 6-dimensional structure obtained by adding a discrete label axis Q to a 5-dimensional hypercube, and describes the combinatorial properties derived from its set-theoretic structure.

In the main body (§1–§8), we assert only the mathematical structure of this model and the consequences derived purely combinatorially therefrom. The correspondence with the Standard Model is presented in §9 as "an example of interpretation," but this is not a claim of this paper. It is merely one example showing how the set-theoretic structure of the model may be mapped to the particle classification of the Standard Model.

On the external assumptions of this model: This paper takes the 5-dimensional hypercube as its starting point. Why 5 dimensions, and why certain axes among the 5 are later treated as "spatial axes," are **external assumptions of this model** and are not justified within the model itself. This dimensional setting is adopted as a correspondence to the empirical fact that the observed spacetime consists of 3 spatial dimensions + 1 time dimension.

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1. Definition of the 6-Dimensional Structure

1.1 Definition

The 6-dimensional structure treated in this paper is defined as follows.

- **Geometric structure**: The combinatorial structure of the 5-dimensional hypercube (vertices, edges, faces, 3-cells, 4-facets, 5-cell body)
- **Label axis**: An 8-valued discrete label axis $Q \in \{0, 1, 2, 3, 4, 5, 6, 7\}$

The vertex coordinates of the 5-dimensional hypercube are denoted by $(x_1, x_2, x_3, x_4, x_5)$, where each coordinate **takes arbitrary real values**. When a specific proposition imposes conditions on coordinate values within the set-theoretic structure, such conditions are declared locally within that proposition.

The 6th axis Q is not a geometric coordinate but a discrete label value associated with each state. Its range is the 8 values $\{0, 1, 2, 3, 4, 5, 6, 7\}$. These 8 values possess a 3-bit structure, encoded as $Q = 4c_1 + 2c_2 + c_3$ ($c_i \in \{0, 1\}$). The addition of the Q axis does not change the combinatorial structure (number of vertices, edges, faces, etc.) of the hypercube.

Open problem. The choice of $2^3 = 8$ values for the range of the Q axis is an external assumption based on consistency with the Standard Model in §9 (an example of interpretation). While the ± 1 symmetry of central projection suggests that the number of Q-axis states should be a power of 2, an internal derivation of why it is $2^3 = 8$ rather than $2^2 = 4$ or $2^4 = 16$ has not yet been achieved.

1.2 Element Counts of the k-Dimensional Hypercube

Let $f_j(k)$ denote the number of j-dimensional faces (j-faces) of the k-dimensional hypercube. We present the element counts starting from low dimensions so that the reader can verify the general formula.

Number of vertices $f_0(k)$: As the combinatorial count distinguishing the 2 states at both ends of each axis:

k	0	1	2	3	4	5
$f_0(k)$	1	2	4	8	16	32

$$f_0(k) = 2^k$$

Number of edges $f_1(k)$: Each edge extends along 1 axis direction and is determined by the position at both ends of the remaining $(k-1)$ axes:

k	0	1	2	3	4	5
$f_1(k)$	0	1	4	12	32	80

$$f_1(k) = k \cdot 2^{k-1}$$

Verification: For $k = 3$, $3 \cdot 4 = 12$ (12 edges of a cube); for $k = 5$, $5 \cdot 16 = 80$.

Number of square faces $f_2(k)$: Faces spanned by 2 axes:

k	0	1	2	3	4	5
$f_2(k)$	0	0	1	6	24	80

$$f_2(k) = \binom{k}{2} \cdot 2^{k-2}$$

Verification: For $k = 3$, $3 \cdot 2 = 6$ (6 faces of a cube); for $k = 5$, $10 \cdot 8 = 80$.

Number of cubic cells (3-cells) $f_3(k)$: Cubic cells spanned by 3 axes:

k	0	1	2	3	4	5
$f_3(k)$	0	0	0	1	8	40

$$f_3(k) = \binom{k}{3} \cdot 2^{k-3}$$

Verification: For $k = 4$, $4 \cdot 2 = 8$; for $k = 5$, $10 \cdot 4 = 40$.

Number of 4-facets $f_4(k)$: 4-dimensional cells spanned by 4 axes:

k	0	1	2	3	4	5
$f_4(k)$	0	0	0	0	1	10

$$f_4(k) = \binom{k}{4} \cdot 2^{k-4}$$

Verification: For $k = 5$, $5 \cdot 2 = 10$.

Number of 5-cell bodies $f_5(k)$: For $k = 5$:

$$f_5(5) = \binom{5}{5} \cdot 2^0 = 1$$

General Formula

Summarizing the above, the number of j-dimensional faces of the k-dimensional hypercube is given by

$$f_j(k) = \binom{k}{j} \cdot 2^{k-j} \qquad (0 \leq j \leq k)$$

This corresponds to choosing j axes from k to span a j-face ($\binom{k}{j}$ ways), and specifying which end of each of the remaining $(k - j)$ axes the face is located at (2^{k-j} ways).

Verification (Euler characteristic): $\sum_{j=0}^{k}(-1)^j f_j(k) = (2-1)^k = 1$, which is consistent with the hypercube being a contractible space.

Values Used in This Paper ($k = 5$)

j	Element	$f_j(5)$
0	Vertices	32
1	Edges	80
2	Square faces	80
3	Cubic cells (3-cells)	40
4	4-facets	10
5	5-cell body	1

1.3 Axial Symmetry and Definition of Set Codes

1.3.1 Complete Symmetry of the 5 Axes **Proposition 1.1 (Complete symmetry of the 5 axes)**. The 5 geometric axes of the 5-dimensional hypercube are all equivalent in their combinatorial structure. Any permutation of axes gives an automorphism of the hypercube, and the 5 axes are indistinguishable with respect to the element counts $f_j(k)$ of §1.2 and subsequent geometric propositions.

Proof. The automorphism group of the combinatorial structure of the k-dimensional hypercube is the hyperoctahedral group B_k, which contains all axis permutations. Therefore the 5 axes are combinatorially completely symmetric. □

1.3.2 Classification of Axes **Definition 1.2 (Notational labels for axes)**. The following notational labels are assigned to the 5 axes x_1, x_2, x_3, x_4, x_5:

- x_1, x_2, x_3 are denoted x, y, z
- x_4 is denoted t
- x_5 is denoted R

By Proposition 1.1, the 5 axes are equivalent in combinatorial structure, so the assignment of labels to axes is merely a definitional choice. These labels are used as notational prerequisites for the definition of set codes (§1.3.3) and the definition of spin (§2).

x, y, z, t, R **are purely notational labels and do not imply any geometric distinction among the axes.** The operation of assigning physical interpretations to each axis is carried out in §9 (an example of interpretation), which is an interpretation from outside the model and is independent of the geometric structure of §1.

1.3.3 Rigorous Definition of Set Codes Each geometric axis component x_i $(i = 1, \ldots, 5)$ is a symbol representing the **range constraint** that the coordinate value along that axis must satisfy. This paper uses the following 5 types of symbols.

Symbol	Meaning	Range of coordinate values
$+$	Positive	$x_i > 0$
$-$	Negative	$x_i < 0$
0	Origin	$x_i = 0$
$\pm$	Nonzero	$x_i \neq 0$ (positive or negative)
$*$	Arbitrary	x_i is any real number

For the Q axis, symbols representing the range of values are used similarly.

Symbol	Meaning	Value of Q
0	Zero label	$Q = 0$
1–7	Specified discrete value	Q is the specified integer value
$\bar{Q}$	Anti-label	$Q \to (Q + 4) \bmod 8$ ($Q = 0$ is invariant)
$*$	Arbitrary	$Q \in \{0, 1, 2, 3, 4, 5, 6, 7\}$ (any value)

The 8 values of Q have a 3-bit structure (c_1, c_2, c_3). The c_1 bit (the most significant bit of Q: whether $Q \geq 4$) and the 2 bits c_2c_3 ($Q \bmod 4$) encode two independent discrete degrees of freedom. The physical interpretation is treated in §9.2.

Remark. Set codes are not coordinate values themselves, but symbols specifying the **conditions** that coordinate values must satisfy. As stated in §1.1, coordinate values in this model can take arbitrary real values, but in specific propositions or set-theoretic structures, the range constraints listed above are imposed on the coordinate values of the relevant axes.

Example. The set $(+, +, 0, 0, 0, 0)$ refers to the collection of states satisfying $x_1 > 0, x_2 > 0, x_3 = 0, x_4 = 0, x_5 = 0, Q = 0$.

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2. Definition of Spin

Definition 2.1 (Spin). For a set (x, y, z, t, R, Q), the spin s is defined as follows.

Fermion/boson determination: Let k denote the number of axes among the 4 axes $xyzt$ whose set code is not 0.

- When $k = 2$: **Fermionic type**
- When $k \neq 2$: **Bosonic type**

Determination of the spin quantum number: The number of axes n contributing to the spin is defined as follows. For the Q axis, only Q label transitions (§3) contribute to the spin; a static Q label value (even if $Q \neq 0$, when no transition is involved) is not counted toward n.

- Fermionic type: n is the number of nonzero elements among $\{R,\ Q\text{-transition}\}$. $s = n+1/2$
- Bosonic type: n is the number of nonzero elements among $\{t,\ R,\ Q\text{-transition}\}$. $s = n$

The possible values of n are $0, 1, 2$ for the fermionic type and $0, 1, 2, 3$ for the bosonic type, so the possible values of spin s in this model are as follows.

n	Boson $s = n$	Fermion $s = n + 1/2$
0	$s = 0$	$s = 1/2$
1	$s = 1$	$s = 3/2$
2	$s = 2$	$s = 5/2$
3	$s = 3$	——

Remark 1. The 4 axes $xyzt$ are completely equivalent in their initial symmetry, and the process by which one axis separates as "temporal" is interpreted as spontaneous symmetry breaking (§9.1). The fermionic determination condition $k = 2$ corresponds to the case where 2 out of 4 equivalent axes are nonzero, and is not restricted to the 3 axes xyz.

Remark 2. This definition is a combinatorial definition internal to this model. The terms "boson" and "fermion" used here are terminology within this model; the correspondence to physical spin angular momentum, statistics, $(-1)^{2s}$ transformation properties, etc. is treated in §9 (an example of interpretation).

Remark 3. The axes counted toward n differ between the fermionic and bosonic types (fermions do not count t). This asymmetry is based on the following geometric motivation. In the fermionic type, exactly 2 out of the 4 axes $xyzt$ are nonzero, and the choice of these 2 axes already carries $\binom{4}{2} = 6$ degrees of freedom. Among these, the combinations including the t axis (3 combinations) and those not including it (3 combinations) have both already participated in the $k = 2$ determination. Therefore the degree of freedom of t has been consumed in determining the fermionic nature, and is reflected in the $1/2$ of the spin quantum number $s = n + 1/2$. As a mathematically equivalent notation, using the number n' of nonzero elements among $\{t, R, Q\text{-transition}\}$, one can also write $s = (n' - 1) + 1/2 = n' - 1/2$ (for fermions with nonzero t, $n' = n + 1$), but this paper adopts the asymmetric form above for clarity of definition.

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3. Enumeration of Q Label Transitions

We consider transitions among the 7 nonzero values $\{1, 2, 3, 4, 5, 6, 7\}$ of the 8 values of the Q axis. Transitions are described by ordered pairs $(Q_{\text{initial}}, Q_{\text{final}})$.

Based on the 3-bit structure (c_1, c_2, c_3), transitions are classified into 2 types.

c_2c_3 **transitions**: Transitions in which the lower 2 bits c_2c_3 change. The ordered pairs among the 3 nonzero values $\{01, 10, 11\}$ of c_2c_3 number $3 \times 3 = 9$ (including 3 diagonal elements).

c_1 **transitions**: Transitions in which only the most significant bit c_1 changes. These are transitions between $c_1 \in \{0, 1\}$, and for each of the 4 values of c_2c_3 $(00, 01, 10, 11)$, there exist 2 directions: $0 \to 1$ and $1 \to 0$. Therefore the number of c_1 transitions is $4 \times 2 = 8$.

Remark. The operation of extracting independent transition states from the 9 c_2c_3 transitions (e.g., removing the linear combination of diagonal elements) depends on representation-theoretic structure and is therefore not performed in the main body. The physical interpretation of the number of independent states is treated in §9.3.

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4. Enumeration of Spin 1 Sets

Spin 1 sets are, by Definition 2.1, bosonic type ($k \neq 2$ among $xyzt$) with $n = 1$. We consider the case where all $xyzt$ axes are 0 ($k = 0$).

The states with exactly 1 nonzero element among $\{t, R, Q\text{-transition}\}$ are classified as follows.

Nonzero axis	Set form	Number of states
t	$(0, 0, 0, \pm, 0, 0)$	2 (2 choices of sign $\pm$)
R	$(0, 0, 0, 0, \pm, 0)$	2 (2 choices of sign $\pm$)
Q-transition (c_2c_3)	$(0, 0, 0, 0, 0, Q_{\text{initial}} \to Q_{\text{final}})$	9 (by §3)
Q-transition (c_1)	$(0, 0, 0, 0, 0, Q_{\text{initial}} \to Q_{\text{final}})$	8 (by §3)

Total: t-fixed 2 states + R-fixed 2 states + c_2c_3 transition 9 states + c_1 transition 8 states = **21 states**.

When only t among $xyzt$ is nonzero ($k = 1$), the state is also bosonic type and includes spin 1 states where t contributes to the count of n.

Remark. The extraction of independent states from the 9 c_2c_3 transition states (reduction $9 \to 8$) and the correspondence to physical gauge bosons are treated in §9.3–§9.4.

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5. Enumeration of Spin 2 Sets

Spin 2 sets are, by Definition 2.1, bosonic type with $n = 2$, i.e., 2 of the 3 elements $\{t, R, Q\text{-transition}\}$ are nonzero, with the number of nonzero axes among $xyzt$ satisfying $k \neq 2$.

For $k = 0$ (all $xyzt$ axes are 0), the combinations of 2 nonzero elements are the following 3 cases.

Nonzero 2 elements	Set form
t, R	$(0, 0, 0, \pm, \pm, 0)$
t, Q-transition	$(0, 0, 0, \pm, 0, Q_i \to Q_j)$
R, Q-transition	$(0, 0, 0, 0, \pm, Q_i \to Q_j)$

Therefore, the number of types of spin 2 sets in this model is **3**.

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6. Enumeration of Spin 3 Sets

Spin 3 sets are, by Definition 2.1, bosonic type with $n = 3$, i.e., all 3 elements $\{t, R, Q\text{-transition}\}$ are nonzero, with the number of nonzero axes among $xyzt$ satisfying $k \neq 2$.

Nonzero 3 elements	Set form
t, R, Q-transition	$(0, 0, 0, \pm, \pm, Q_i \to Q_j)$

Therefore, the number of types of spin 3 sets in this model is **1**.

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7. Sign Product of Nonzero Axes

Definition 7.1 (Sign product). For a bosonic set, when the axes among t, R whose set code is not 0 each take a sign of $+$ or $-$, the product of these signs is defined as the **sign product** P.

$$P = (-1)^{|\{i \in \{t,R\} : \sigma_i < 0\}|}$$

That is, $P = +1$ if the number of axes among t, R with negative sign is even, and $P = -1$ if odd.

Proposition 7.1 (Classification of P for each spin type). The possible values of P for each bosonic type are as follows.

Spin type	Nonzero $\{t, R\}$	Possible values of P
$s = 0$	None	$P = +1$ (unique)
$s = 1$, t type	t only	$P = +1$ or -1
$s = 1$, R type	R only	$P = +1$ or -1
$s = 1$, Q-transition type	None	$P = +1$ (unique)
$s = 2$, tR type	Both t, R	$P = +1$ or -1
$s = 2$, tQ type	t only	$P = +1$ or -1
$s = 2$, RQ type	R only	$P = +1$ or -1
$s = 3$, tRQ type	Both t, R	$P = +1$ or -1

Proof. When both t, R are zero, $P = +1$ (0 negative axes). When only 1 is nonzero, $\sigma_i = \pm$ gives $P = \pm 1$. When both are nonzero, the 4 configurations $(+)(+) \to P = +1$, $(+)(-) \to P = -1$, $(-)(+) \to P = -1$, $(-)(-) \to P = +1$ are possible, giving $P = \pm 1$. $\square$

The physical interpretation of P (correspondence with the directionality of forces) is treated in §9 (an example of interpretation).

Remark. The orientation structure of signed areas $M(\sigma)$ for square faces containing the R axis, and the rigorous definition of conserved quantities based thereon, exceed the scope of this paper and will be treated in a separate paper. Interpretive speculation on the relation between $M(\sigma)$ and the mass metric is given in §9.11.1.

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8. Conclusion of the Main Body

In this paper, we defined a 6-dimensional structure obtained by adding an 8-valued discrete label axis Q to a 5-dimensional hypercube, and derived the following combinatorial properties from its set-theoretic structure.

1. **Element counts of the hypercube** (§1.2): The number of j-dimensional faces of the k-dimensional hypercube is $f_j(k) = \binom{k}{j} \cdot 2^{k-j}$. For $k = 5$: 32 vertices, 80 edges, 80 square faces, 40 cubic cells, 10 hypercubes.
2. **Complete symmetry of the 5 axes** (Proposition 1.1): The 5 geometric axes are all equivalent in combinatorial structure, and any axis permutation gives an automorphism of the hypercube.
3. **Notational labels for axes** (Definition 1.2): The notational labels x, y, z, t, R are assigned to the 5 axes.
4. **Rigorous definition of set codes** (§1.3.3): Range constraints on each axis are expressed using the 5 symbols $+, -, 0, \pm, *$. The Q axis has 8 values (3-bit structure).
5. **Definition of spin** (Definition 2.1): The fermionic type ($k = 2$) and bosonic type ($k \neq 2$) are determined by the number k of nonzero axes among the 4 axes $xyzt$, and the spin is defined as $s = n + 1/2$ or $s = n$ from the number n of nonzero elements among $\{R, Q\text{-transition}\}$ (fermionic type) or $\{t, R, Q\text{-transition}\}$ (bosonic type).
6. **Enumeration of Q label transitions** (§3): c_2c_3 transitions (9 cases) and c_1 transitions (8 cases) based on the 3-bit structure.
7. **Enumeration of spin 1 sets** (§4): t-fixed 2 states + R-fixed 2 states + c_2c_3 transition 9 states + c_1 transition 8 states = 21 states.
8. **3 types of spin 2 sets** (§5): tR, tQ, RQ — 3 types.
9. **1 type of spin 3 sets** (§6): tRQ — 1 type.
10. **Sign product** (Definition 7.1, Proposition 7.1): The sign product $P = (-1)^{|\{i:\sigma_i<0\}|}$ is defined from the number of negative-sign axes among t, R, and the possible values of P (unique or both values) are classified for each spin type.

All of these are results derived purely from combinatorics and geometry under the external assumptions of this model (adoption of the 5-dimensional hypercube, addition of the Q axis).

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9. An Example of Interpretation: Correspondence with the Standard Model

The content of this section is not a claim of this paper but an example of interpretation. It does not attempt to physically derive the Standard Model from this model. This section presents one example of how the 6-dimensional structure defined and derived in §1–§8 and its mathematical properties may be mapped to the particle classification of the Standard Model.

The following correspondences are not justified from within this model and are **choices of interpretation from outside**. If an alternative mapping becomes available in the future, it can be added in parallel with this section.

9.1 Assignment of Physical Labels

As stated in §1.3.2, the 5 axes are all equivalent in combinatorial structure; however, for the purpose of correspondence with the Standard Model, the following labels are assigned.

- $x_1, x_2, x_3 \to xyz$: 3-dimensional spatial axes
- $x_4 \to t$: Time axis
- $x_5 \to R$: Spatial scale axis (an axis connecting the subjective coordinate system with the spatial scale of the 6-dimensional structure)
- Q: Gauge symmetry axis (discrete encoding of $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$)

Initial symmetry of $xyzt$: By Proposition 1.1, the 4 axes $xyzt$ are completely equivalent in the initial state. The process by which one of these axes separates as "time" in observation

is interpreted as spontaneous symmetry breaking. This symmetry breaking gives rise to the distinction between charged particles (t axis nonzero) and neutral particles (t axis zero).

9.2 Correspondence Between the 8 Values of the Q Axis and Gauge Symmetry

The 8 values of the Q axis consist of 3 bits (c_1, c_2, c_3). These 3 bits encode 2 gauge degrees of freedom.

Q	(c_1, c_2, c_3)	c_2c_3 (color)	c_1 (weak isospin)	Correspondence
0	(0, 0, 0)	00 (colorless)	0 (down)	Lepton (neutral)
1	(0, 0, 1)	01 (color 1)	0 (down)	Quark (down-type, color 1)
2	(0, 1, 0)	10 (color 2)	0 (down)	Quark (down-type, color 2)
3	(0, 1, 1)	11 (color 3)	0 (down)	Quark (down-type, color 3)
4	(1, 0, 0)	00 (colorless)	1 (up)	Lepton (charged)
5	(1, 0, 1)	01 (color 1)	1 (up)	Quark (up-type, color 1)
6	(1, 1, 0)	10 (color 2)	1 (up)	Quark (up-type, color 2)
7	(1, 1, 1)	11 (color 3)	1 (up)	Quark (up-type, color 3)

Particles with $c_2c_3 = 00$ correspond to leptons (no color charge), and particles with $c_2c_3 \neq 00$ correspond to quarks (with color charge). c_1 encodes the up/down weak isospin.

Antiparticles correspond to the transformation $Q \to (Q + 4) \bmod 8$ (inversion of the c_1 bit), accompanied by sign inversion of the spatial axes.

Scalar mapping functions for the Q axis. The following mapping functions are defined to extract physical quantity labels from the discrete label on the Q axis. These functions are abstract conversion rules for reading the discrete label Q as scalar values used in the Standard Model; they do not assert the existence of any physical entity or any underlying dynamical mechanism.

Definition 9.1 (Charge mapping f_Q^{charge}). Define $f_Q^{\text{charge}} : \{0, 1, \ldots, 7\} \to \mathbb{Q}$ as follows:

Q	$f_Q^{\text{charge}}(Q)$
0	0
1,2,3	$-1/3$
4	-1
5,6,7	$+2/3$

$f_Q^{\text{charge}}(0) = 0$ corresponds to the state orthogonal to the Q axis ($c_2c_3 = 00$, $c_1 = 0$), signifying the absence of electric charge.

Definition 9.2 (Color charge mapping f_Q^{color}). Define $f_Q^{\text{color}} : \{0, 1, \ldots, 7\} \to \{0, 1, 2, 3\}$ by $f_Q^{\text{color}}(Q) = Q \bmod 4$ (i.e., the value $2c_2 + c_3$ of c_2c_3). $f_Q^{\text{color}}(Q) = 0$ corresponds to colorless (lepton), and $f_Q^{\text{color}}(Q) \in \{1, 2, 3\}$ corresponds to the 3 color charges (quark).

Definition 9.3 (Isospin mapping f_Q^{iso}). Define $f_Q^{\text{iso}} : \{0, 1, \ldots, 7\} \to \{0, 1\}$ by $f_Q^{\text{iso}}(Q) = c_1 = \lfloor Q/4 \rfloor$. $c_1 = 0$ corresponds to down-type, and $c_1 = 1$ corresponds to up-type.

Conserved quantity mappings from vectors. The following mapping functions are declared to return conserved quantities from the 6-dimensional sign vector (x, y, z, t, R, Q) and the curvature radius R_1 ($R_1 \geq 0$) of the subjective space. These are abstract functions expressing the hypothesis that "some conserved quantity should be determined from the sign vector"; the functional form is undetermined. They do not assert that concrete function definitions are achievable, nor do they presuppose the physical existence of corresponding conserved quantities.

Definition 9.4 (Rest mass mapping f_m^{rest}). $f_m^{\text{rest}} : (x, y, z, t, R, Q) \to \mathbb{R}_{\geq 0}$ is a mapping that returns a scalar quantity corresponding to the rest mass from the sign vector. f_m^{rest} may depend on f_Q^{charge}, f_Q^{color}, f_Q^{iso}, and the axis scale values v_i.

Definition 9.5 (Curvature energy mapping f_m^{curv}). $f_m^{\text{curv}} : (x, y, z, t, R, Q, R_1) \to \mathbb{R}_{\geq 0}$ is a mapping that returns a scalar quantity corresponding to curvature-derived energy (acceleration energy) from the sign vector and the curvature radius R_1 of the subjective space. When $R_1 = 0$, $f_m^{\text{curv}} = 0$.

9.3 Correspondence to the 8 Gluon States

Of the 9 c_2c_3 transitions enumerated in §3, removing the linear combination of one diagonal element (the state proportional to the color singlet $r\bar{r} + g\bar{g} + b\bar{b}$ in SU(3)), the **8 independent states are mapped to gluons** $g_1, \ldots, g_8$.

This reduction $9 - 1 = 8$ originates from the adjoint representation of the Lie algebra $\mathfrak{su}(3)$ of SU(3) (the decomposition $3 \otimes \bar{3} = 1 \oplus 8$). While 9 transition pairs are derived from the combinatorics of this model (§3), the operation of removing 1 state to obtain 8 depends on the representation-theoretic interpretation. The non-abelian structure constants f^{abc} of SU(3) are not derived from this model.

9.4 Correspondence to Gauge Bosons

Among the spin 1 sets enumerated in §4, the correspondence to Standard Model gauge bosons is shown below.

Fixed axis	Correspondence	Number of states
t	W^+, W^-	2
R	γ, Z^0	2
Q (c_2c_3 transition, §9.3)	$g_1, \ldots, g_8$	8
Total		**12**

The 2 states of the W boson ($t = +1, -1$) correspond to the positive and negative electric charges, while γ and Z^0 correspond to the sign of the R axis (+ vs −). The total of 12 matches the number of gauge boson states in the Standard Model: $\gamma + g \times 8 + W^+ + W^- + Z^0 = 12$.

The 8 states of c_1 transitions (§3) can be interpreted as another aspect of weak isospin transitions mediated by W bosons, but their counting as independent particle states is treated in §9.10.

9.5 Interpretation of Spin 2 Sets

The following correspondences are assigned to the 3 types of spin 2 sets enumerated in §5.

Type	Fixed elements	Correspondence
tR type	t, R	**Graviton** (correspondence within the Standard Model + gravity framework)
$t \cdot Q$-transition type	t, Q-transition	No known particle in the Standard Model
$R \cdot Q$-transition type	R, Q-transition	No known particle in the Standard Model

The tR type is consistent with the properties of gravity (attraction only). The tQ and RQ types are states whose existence is derived from the combinatorial structure of this model, but they have no counterpart in the current Standard Model.

9.6 Interpretation of Spin 3 Sets

The single type of spin 3 set (§6), the tRQ-transition type, has no known corresponding particle in the Standard Model.

9.7 Correspondence Between Sign Product and Directionality of Forces

The following physical interpretation is assigned to the sign product P defined in §7.

- $P = +1$: The force is unidirectional (**attraction only**)
- $P = -1$: The force is bidirectional (**both attraction and repulsion**)

By Proposition 7.1, the tR type of spin 2 admits both $P = +1$ and $P = -1$ configurations. Interpreting the $P = +1$ configuration (same sign for t, R) as mediating gravity yields attraction only, which is consistent with the properties of gravity. What the $P = -1$ configuration (opposite signs for t, R) corresponds to physically is unresolved, and remains as a future problem: either (a) a configuration not physically realized, or (b) a correspondence to an unknown phenomenon.

Similarly, the t and R types of spin 1 admit both $P = \pm 1$ configurations, consistent with forces having both attraction and repulsion (electromagnetic force, weak force). The Q-transition type of spin 1 has $P = +1$ (unique), suggesting a relationship with the confinement property of gluons.

9.8 Range of the R Axis and Coupling Strength of Forces

Range of the R axis: In prior work [W5], the number-theoretic conditions of central projection in discrete space lead to the discretization $R = \pm n^2 \cdot L_0$ ($n \in \mathbb{N}$, L_0 is the unit length). The R axis connects the subjective coordinate system with the spatial scale of the 6-dimensional structure, and corresponds to the unboundedness of the mass spectrum.

Interpretation of the correspondence between each force and the axes of this model:

Force	Mediator	Involved axes	Coupling characteristics
Strong force	g_1–g_8	Q axis (c_2c_3 color transitions, 8 channels, §9.3)	Only between particles with color charge
Weak force	$W^{\pm}, Z^0$	t axis + Q axis (c_1 transitions)	All fermions
Electromagnetic	γ	R axis	Between particles with nonzero electric charge

Force	Mediator	Involved axes	Coupling characteristics
Gravity	G	t, R, both axes nonzero ($n = 2$)	All particles (Q-independent)

9.9 Table A: Correspondence Between Confirmed Particles and 6-Dimensional Sets

This table shows particles only. Antifermions are obtained by sign inversion of spatial axes and $Q \to (Q + 4) \bmod 8$, yielding 24 states: 6 antileptons + 18 antiquarks. For bosons, particle-antiparticle pairs such as W^+/W^- are explicitly listed in the table; γ, Z^0, H^0, G, and gluons are self-conjugate.

Bosons:

Particle	Symbol	s	Set (x, y, z, t, R, Q)	n	Mass (MeV)
Higgs	H^0	0	$(0, 0, +, 0, 0, 0)$	0	125,250
W boson	W^+	1	$(0, 0, 0, +, 0, 0)$	1	80,379
W boson	W^-	1	$(0, 0, 0, -, 0, 0)$	1	80,379
Photon	γ	1	$(0, 0, 0, 0, +, 0)$	1	0
Z boson	Z^0	1	$(0, 0, 0, 0, -, 0)$	1	91,188
Gluon	g_1–g_8	1	$(0, 0, 0, 0, 0, Q_i \to Q_j)$	1	0
Graviton	G	2	$(0, 0, 0, \pm, \pm, 0)$	2	0

Verification: Higgs: z axis only nonzero, $k = 1$ (bosonic type), $n = 0$, $s = 0$ ✓. W: $k = 1$ (bosonic type), t nonzero so $n = 1$, $s = 1$ ✓. Photon: $k = 0$ (bosonic type), R nonzero so $n = 1$, $s = 1$ ✓. Gluon: $k = 0$ (bosonic type), Q-transition so $n = 1$, $s = 1$ ✓. Graviton: $k = 0$ (bosonic type), t, R nonzero so $n = 2$, $s = 2$ ✓.

Selection of the z axis for the Higgs: The nonzero axis of the Higgs boson involves an interpretive degree of freedom in choosing 1 axis from the 4 axes $xyzt$. For consistency with the generation labels (generation 1: x, generation 2: y, generation 3: z), the z axis (3rd generation) is selected. This choice is consistent with the fact that 3rd-generation fermions (top: 172,760 MeV) have the largest mass, provided the coupling strength depends on the overlap along shared spatial axes (see §9.11.7).

Leptons:

Particle	Symbol	s	Generation	Set	Q	Mass (MeV)
Electron	e^-	1/2	1	$(+, 0, 0, +, 0, 4)$	4	0.511
Electron neutrino	ν_e	1/2	1	$(+, +, 0, 0, 0, 0)$	0	$< 10^{-6}$
Muon	μ^-	1/2	2	$(0, +, 0, +, 0, 4)$	4	105.66
Muon neutrino	ν_μ	1/2	2	$(0, +, +, 0, 0, 0)$	0	< 0.17
Tau	τ^-	1/2	3	$(0, 0, +, +, 0, 4)$	4	1,776.9
Tau neutrino	ν_τ	1/2	3	$(+, 0, +, 0, 0, 0)$	0	< 15.5

Verification: Electron: x, t are nonzero among $xyzt$, so $k = 2$ (fermionic type); $\{R, Q\text{-transition}\}$ are both not applicable ($R = 0$, $Q = 4$ is a static label, not a transition), so $n = 0$, $s = 0 + 1/2 = 1/2$ ✓. Neutrino: x, y are nonzero, so $k = 2$ (fermionic type), $n = 0$, $s = 1/2$ ✓.

Characteristics of charged leptons: The t axis is nonzero, with $Q = 4$ ($c_1 = 1, c_2c_3 = 00$; weak isospin up, colorless). Neutrinos have $t = 0$, $Q = 0$, with only 2 spatial axes among $xyzt$ being nonzero.

Distinction of generations: For charged leptons, 1 nonzero spatial axis and the t axis satisfy $k = 2$; for neutrinos, 2 spatial axes satisfy $k = 2$. The generation label is determined by the nonzero spatial axes (generation 1: x, generation 2: y, generation 3: z).

Generation correspondence rule: In the lepton doublet $\{\ell_n^-, \nu_n\}$ of the same generation, one of the 2 nonzero spatial axes of ν_n coincides with the nonzero spatial axis of ℓ_n^-. The choice of the remaining axis is an interpretive degree of freedom; this paper adopts a cyclic assignment (generation 1: xy, generation 2: yz, generation 3: xz).

Antileptons correspond to sign inversion of spatial axes and $Q \to (Q+4) \bmod 8$ (e^+: $Q = 4 \to 0$).

Quarks:

Particle	Symbol	s	Generation	Set	Q	Mass (MeV)
Up	u	1/2	1	$(+, 0, 0, +, 0, 5/6/7)$	5,6,7	2.16
Down	d	1/2	1	$(+, 0, 0, -, 0, 1/2/3)$	1,2,3	4.67
Charm	c	1/2	2	$(0, +, 0, +, 0, 5/6/7)$	5,6,7	1,270
Strange	s	1/2	2	$(0, +, 0, -, 0, 1/2/3)$	1,2,3	93.4
Top	t	1/2	3	$(0, 0, +, +, 0, 5/6/7)$	5,6,7	172,760
Bottom	b	1/2	3	$(0, 0, +, -, 0, 1/2/3)$	1,2,3	4,180

Verification: Up quark: x, t are nonzero among $xyzt$, so $k = 2$ (fermionic type); $Q = 5$ is a static label (not a transition), so $n = 0$, $s = 0 + 1/2 = 1/2$ ✓.

Characteristics of quarks: All quarks have the t axis nonzero. Up-type quarks (u, c, t) have $t > 0$ with $Q \in \{5, 6, 7\}$ ($c_1 = 1$; weak isospin up, colored). Down-type quarks (d, s, b) have $t < 0$ with $Q \in \{1, 2, 3\}$ ($c_1 = 0$; weak isospin down, colored). The sign of the t axis reflects the sign structure of the electric charge.

Each quark has 3 color states corresponding to the color values of Q ($c_2c_3 \in \{01, 10, 11\}$). Antiquarks correspond to sign inversion of spatial axes and $Q \to (Q + 4) \bmod 8$.

Summary:

Classification	s	Q	States derived in this model	SM correspondence
Higgs	0	0	1	1
Gauge boson (t type)	1	0	2	W^+, W^-
Gauge boson (R type)	1	0	2	γ, Z^0
Gluon (Q-transition type)	1	Transition	8	$g_1, \ldots, g_8$
Graviton (tR type)	2	0	1	G (beyond SM)
Lepton	1/2	0 or 4	6	6
Antilepton	1/2	0 or 4	6	6
Quark	1/2	1,2,3 or 5,6,7	18	18
Antiquark	1/2	1,2,3 or 5,6,7	18	18

Classification	s	Q	States derived in this model	SM correspondence
Subtotal			**62**	**62 (61 + G)**

The 62 states map to 61 Standard Model states + 1 graviton. The excess color singlet state in the previous model ($Q = 4$ values) has been resolved by the extension to $Q = 8$ values.

In addition to the above, combinations with $s = 3/2, 2, 5/2, 3$ (Table B, §9.10) are derived from this model, but no known particles in the Standard Model correspond to them.

9.10 Table B: Combinations with No Counterpart in the Standard Model

Listed below are the states derived from the combinatorial structure of this model for which no known particle in the Standard Model exists. These are combinatorial consequences within this model and do not assert the existence of undiscovered particles.

Bosons (number of nonzero axes among $xyzt$: $k \neq 2$):

Type	s	n	Set form	Number of states
tQ type	2	2	$(0, 0, 0, \pm, 0, Q_i \to Q_j)$	$2 \times N_Q$
RQ type	2	2	$(0, 0, 0, 0, \pm, Q_i \to Q_j)$	$2 \times N_Q$
tRQ type	3	3	$(0, 0, 0, \pm, \pm, Q_i \to Q_j)$	$4 \times N_Q$

Here N_Q is the number of Q transition states: c_2c_3 transition 9 states + c_1 transition 8 states = 17 (§3). Under the interpretation of §9.3 where c_2c_3 transitions are reduced $9 \to 8$, $N_Q = 16$.

Fermions (number of nonzero axes among $xyzt$: $k = 2$, excluding combinations included in Table A):

With the extension to $Q = 8$ values, the number of fermionic-type combinations increases substantially. Choosing 2 axes from $xyzt$: $\binom{4}{2} = 6$ ways $\times$ signs of each axis $2^2 = 4$ ways $\times$ 8 Q values $= 6 \times 4 \times 8 = 192$ fermionic states exist. Of these, the 48 states mapped in Table A (6 leptons + 6 antileptons + 18 quarks + 18 antiquarks) are subtracted, leaving the remainder as unassigned states.

Higher-spin fermions (with R axis or Q-transition nonzero, $n \geq 1$):

Type	s	n	Additional nonzero axis
R type	3/2	1	R
Q-transition type	3/2	1	Q (transition)
RQ type	5/2	2	R, Q (transition)

These higher-spin fermions are states where, in addition to 2 nonzero axes among $xyzt$, R or Q-transitions are also nonzero. They have an origin distinct from the partner particles of supersymmetric theories.

9.11 New Structural Consequences and Open Problems in the Interpretation Example

9.11.1 Initial symmetry and value structure of the $xyzt$ axes

Since the 4 axes $xyzt$ are completely equivalent in their initial symmetry, each axis is expected to take values of the same kind (positive integers). However, depending on the number of nonzero axes, there exist 4 types of values v_1, v_2, v_3, v_4 (for the cases of 1, 2, 3, and 4 nonzero axes, respectively), and these are generally distinct.

What is structurally established is only that $v_1 \neq v_2 \neq v_3 \neq v_4$ is permitted. Everything below is **interpretive speculation** and is not derived from within this model.

Relation to dimensional analysis: In the central projection model of prior work [W5], spatial length L and mass M have the dimensional relation $M = L$ (in natural units). Applying this relation to this model, the sum of absolute values of coordinate values along the $xyzt$ axes may be related to the mass metric $M(\sigma)$. That is, for a set $\sigma = (x, y, z, t, R, Q)$,

$$M(\sigma) \sim \sum_{i \in \{x,y,z,t\}} |x_i|$$

may serve as the origin of mass. However, how the R axis (spatial scale axis) and Q axis (gauge symmetry axis) contribute to this metric is undetermined, and a more general form including R, Q,

$$M(\sigma) \sim f\Big(\sum |x_i|,\ R,\ Q\Big)$$

cannot be excluded.

Speculation that $v_1 \gg v_2$: While the mass of the Higgs boson ($k = 1$, z axis only nonzero, v_1) is 125,250 MeV, the majority of fermions ($k = 2$, 2 nonzero axes, v_2) are far lighter (electron 0.511 MeV, neutrinos $< 10^{-6}$ MeV). If the above dimensional analysis speculation is correct, then $v_1 \gg v_2$ must hold, suggesting that the coordinate value for the case of only 1 nonzero axis is substantially larger than that for 2 nonzero axes.

Inter-generation mass hierarchy: The assignment of the Higgs boson's nonzero axis to z (the 3rd-generation axis) provides a structural indication for the mass differences among generations within the v_2 range. 3rd-generation fermions (z axis nonzero) directly share a spatial axis with the Higgs standing wave and therefore have the strongest coupling, while 1st-generation (x axis) and 2nd-generation (y axis) fermions lack a shared axis and are limited to indirect coupling. This is consistent with the empirical hierarchy $m_3 \gg m_{1,2}$ (top 172,760 MeV $\gg$ charm 1,270 MeV, up 2.16 MeV). However, since the x and y axes are locally equivalent under SO(3) symmetry, the difference $m_2 \neq m_1$ cannot be derived within this framework. The quantitative derivation of mass ratios remains an open problem.

Relation to the problem of undiscovered $k = 3, 4$ states: States with 3 or 4 nonzero axes among $xyzt$ ($k = 3, 4$) are derived from this model as bosonic type ($k \neq 2$), but no known particles in Table A correspond to them. If the values v_3, v_4 corresponding to these states are extremely small (or near zero), the energy scale of the corresponding particles may exceed the current experimental reach. Conversely, if v_3, v_4 are extremely large, the corresponding particles may not be physically realized due to stability issues. In either case, the fact that $k = 3, 4$ states remain undiscovered is likely closely related to the value structure of the $xyzt$ axes.

9.11.2 The $Q = 0$ constraint and free particles

$Q = 0$ signifies neutrality on the gauge symmetry axis. As an interpretation, the constraint that only particles with $Q = 0$ are observable as free particles is conceivable. This can be interpreted as the geometric origin of quark confinement (particles with $Q \neq 0$ are not observed individually).

9.11.3 Predictions of undiscovered particles

With the extension of the Q axis from 4 to 8 values, the number of combinatorial states derived from this model has increased substantially. These combinations beyond the 62 states of Table A are to be examined in future research as either (a) predictions of undiscovered particles, (b) physically unrealized combinations (existence of selection rules), or (c) limitations of this interpretation.

In particular, higher-spin fermions ($s = 3/2, 5/2$) and additional bosons (tQ type, RQ type spin 2; tRQ type spin 3) have no counterparts among known particles.

9.11.4 Derivation of mass ratios

Deriving the known particle mass ratios from the set-theoretic structure of this model is a task for future work. The specific ratios of the 4 types of values v_1, v_2, v_3, v_4 for the $xyzt$ axes (§9.11.1), and the form of the contribution of the R and Q axes to the mass metric $M(\sigma)$, may be key to this problem. In particular, if the relation $v_1 \gg v_2$ (§9.11.1) is confirmed, a geometric explanation for the mass hierarchy between the Higgs boson and fermions would be obtained.

9.11.5 Derivation of gauge groups

A correspondence is suggested between the 8 values (3-bit structure) of the Q axis and $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$, and between the continuous scale of the R axis and $\mathrm{U}(1)_Y$, but the rigorous mechanism for the transition from discrete structure to continuous gauge groups remains unproven.

9.11.6 CKM matrix and PMNS matrix

How generation mixing (CKM matrix, PMNS matrix) is derived as the breaking of S_3 symmetry of the xyz axes is an open problem.

9.11.7 Higgs mechanism and generation mass hierarchy

The Higgs boson corresponds to only the z axis being nonzero ($k = 1$, Table A: $(0, 0, +, 0, 0, 0)$). The z axis is the label axis for the 3rd generation (top, bottom, tau), and this assignment has the following consistency:

- Since the coupling strength to the Higgs depends on the overlap along shared spatial axes, 3rd-generation fermions with nonzero z axis have the strongest coupling and acquire the largest mass (top: 172,760 MeV).
- 1st-generation (x axis) and 2nd-generation (y axis) fermions do not share the Higgs's primary oscillation axis, resulting in weaker coupling and lighter mass.

How spontaneous symmetry breaking by the Higgs field is understood within this framework remains a task for future work, but the selection of the z axis provides a geometric explanation for the inter-generation mass hierarchy ($m_3 \gg m_{1,2}$).

9.11.8 Complete derivation of charge values

The sign of the electric charge can be read from the sign of the t axis, and the presence/absence of color charge and the absolute value of electric charge (1 vs 1/3) from c_2c_3 of the Q axis, but the complete derivation of the asymmetry between $+2/3$ and $-1/3$ is a task for future work.

9.12 Conclusion of the Interpretation Example

This section presented one example of mapping the mathematical properties of the 6-dimensional structure defined and derived in §1–§8 to the particle classification of the Standard Model.

1. The 8 values (3-bit structure) of the Q axis discretely encode the $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$ gauge symmetry.

2. From the initial symmetry of the 4 axes $xyzt$, the definition of spin arises naturally (fermion: $k = 2$; boson: $k \neq 2$).
3. 62 states can be mapped to known particles of the Standard Model (61 + graviton).
4. Of the 9 c_2c_3 transitions, the 8 states remaining after removing the color singlet correspond to gluons (the reduction $9 \to 8$ depends on SU(3) representation theory).
5. The 12 states of spin 1 sets correspond to the gauge bosons of the Standard Model.
6. Charged fermions have the t axis nonzero, and the sign of the electric charge corresponds to the sign of t.
7. The $Q = 0$ constraint can be interpreted as the geometric origin of quark confinement.
8. The value structure of the $xyzt$ axes (4 types of positive integer values) may contain the origin of the mass hierarchy.
9. Higher-spin states and surplus states accompanying the extension to $Q = 8$ values suggest either predictions of undiscovered particles or the existence of selection rules.

The derivation of specific mass values, charge values, the non-abelian structure of gauge groups, and generation mixing matrices are tasks for future work (§9.11).

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